

Smart Construction Connect

AI-Powered Construction Matchmaking Platform — FYP Documentation (Final)

Environment: Localhost only | **Data Layer:** JSON files | **No auth/deployment scope**

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1. Project Overview

Smart Construction Connect is an AI-powered web platform that intelligently connects four key

stakeholders in the construction industry:

Stakeholder	Role
 Client	Homeowners or buyers looking to build or renovate
 Construction Company	Service providers seeking verified project leads
 Material Supplier	Vendors selling construction materials
 Admin	Platform guardian ensuring trust and quality

The platform replaces fragmented, manual processes — word-of-mouth referrals, cold calls, physical visits — with a centralized, intelligent, and transparent matchmaking system powered by AI.

Tech Stack

Layer	Technology
Frontend	React.js (Vite), Tailwind CSS, Framer Motion
State Management	Zustand
Routing	React Router v6
Data Layer	Local JSON files (no database for FYP)
AI Chat UI	Custom UI + Claude/OpenAI API (Phase 3)
Icons	Lucide React

2. Problem Statement

The construction industry in Pakistan (and developing markets broadly) faces:

- No centralized directory of verified, rated construction companies
- No price transparency — clients are often overcharged
- Supplier discovery is manual — companies don't know where to source materials efficiently
- No accountability — bad actors (fraudulent companies, fake reviews) go unchecked
- Language barrier — most platforms are English-only, excluding Urdu-speaking clients

Smart Construction Connect directly addresses all five of these problems.

Key Statistics (Pakistan Context)

- 60%+ of construction disputes arise from mismatched expectations
 - Clients typically contact 5-10 companies manually before deciding
 - Material procurement is largely offline, causing 20-30% cost inefficiencies
 - No existing platform serves this space in Pakistan at scale
-

3. Vision & Goals

Vision

To become the most trusted AI-driven construction ecosystem in Pakistan — where clients find verified companies, companies get quality leads, and suppliers grow their business efficiently.

FYP Goals

- Build a complete, role-based frontend with realistic JSON dummy data
- Demonstrate a working AI chatbot UI for premium client matchmaking
- Design a scalable system architecture that can evolve into a real product
- Showcase admin governance tools for platform trust

Long-Term Goals (Post-FYP)

- Real backend with live company/supplier database (PostgreSQL)
 - Payments integration (Stripe / JazzCash / EasyPaisa)
 - Mobile app (React Native)
 - Cloud deployment (Vercel + AWS / Railway)
-

4. User Personas

Persona 1: Raza (Client — Normal Mode)

- **Age:** 38, Government employee
- **Goal:** Build a 10-Marla house in DHA Lahore on a 1.5 Crore budget
- **Pain:** Doesn't know which company is genuine; has been misquoted before
- **Tech literacy:** Medium — uses WhatsApp, browses Facebook
- **Needs:** Simple filtering, company ratings, compare 2-3 companies side-by-side

 Persona 2: Hania (Client — Premium Mode)

- **Age:** 29, IT professional
- **Goal:** Custom contemporary home; wants AI assistance in shortlisting
- **Pain:** Too busy to manually research; wants smart recommendations
- **Tech literacy:** High — comfortable with AI chatbots
- **Needs:** AI assistant in English/Urdu, explainable recommendations, fast decision-making

 Persona 3: M. Shahid (Construction Company Owner)

- **Age:** 45, runs a mid-size company
- **Goal:** Get 3-4 new verified project leads per month
- **Pain:** Marketing is word-of-mouth; no digital portfolio
- **Needs:** Professional profile page, lead management, material sourcing

 Persona 4: Tariq (Material Supplier)

- **Age:** 52, brick and cement wholesale dealer
- **Goal:** Connect directly with construction companies; reduce idle inventory
- **Pain:** Depends on brokers; no digital storefront
- **Needs:** Product listings, inventory management, order tracking

 Persona 5: Admin

- **Role:** Platform operator
- **Goal:** Maintain platform integrity and trust
- **Needs:** Full control over approvals, bans, content moderation, analytics

5. User Roles & Permissions

Feature	Client (Normal)	Client (Premium)	Company	Supplier	Admin
Browse Companies	✓	✓	✗	✗	✓
Filter & Search	✓	✓	✗	✗	✓
Compare Companies	✓ (manual)	✓	✗	✗	✗
AI Chat Assistant	✗	✓	✗	✗	✗
AI Recommendations	✗	✓	✗	✗	✗

Feature	Client (Normal)	Client (Premium)	Company	Supplier	Admin
Request Quotes	✓	✓	✗	✗	✗
Manage Projects	✗	✗	✓	✗	✗
Post Materials	✗	✗	✗	✓	✗
Write Reviews	✓	✓	✗	✗	✗
Approve Companies	✗	✗	✗	✗	✓
Ban Users	✗	✗	✗	✗	✓
View Audit Logs	✗	✗	✗	✗	✓
Moderate Reviews	✗	✗	✗	✗	✓

Note on Onboarding: Companies and Suppliers enter a **"Pending"** state after signup. They see a read-only dashboard and cannot operate until Admin approves them. For FYP, this state is simulated via JSON flags.

6. Client Experience

6.1 Normal Mode

Flow: Dashboard → Browse Companies → Filter → Compare → Request Quote

Features:

- Search and browse verified construction companies
- Filter by: budget range, rating, location (city/area), specialization, availability, verified-only toggle
- Sort by: Best Match, Highest Rated, Lowest Price, Most Reviews
- Side-by-side comparison of up to 3 companies (price range, rating, project types, location, completion time, verified status)
- Request a quote: fill Budget, Area, Timeline, Description → company notified
- View company profile: portfolio, reviews, team size, past projects
- Save/bookmark companies to a wishlist

6.2 Premium Mode — AI Matchmaking

Flow: Activate Premium → AI Chat (Sara/Marcus) → Requirements Collected → Live Analysis Panel → Top 5 Recommendations

Split-Screen UI Layout:

AI CHAT (LEFT)	LIVE PANEL (RIGHT)
Sara: "Assalam o Alaikum! Let's find your perfect construction company. What is your budget?"	📋 Requirements Summary Budget: 1.5 Crore Area: 10 Marla Style: Modern Location: DHA Lahore Timeline: 6 months
User: "1.5 crore"	
Sara: "Great! What area are you building on?"	⚙️ AI Processing Logs ✓ Budget analyzed ✓ Location matched ⌚ Calculating style score...
[Text input] [Send]	
[🎤 Voice] [اردو]	🏆 Top Recommendations 1. BuildPro Lahore 94% ✓ 2. UrbanCraft Co. 88% 3. Noor Builders 82%

AI Agent Personas (user selects on first launch):

- **Sara** — Female, friendly, warm tone (default)
- **Marcus** — Male, professional, formal tone

Data Sara/Marcus Collects:

Field	Example
Budget	"1.5 crore", "50 lakh"
Area	"10 marla", "1 kanal"
Location	"DHA Lahore", "Bahria Town Karachi"
Construction Style	"Modern", "Traditional", "Mixed"
Timeline	"6 months", "as soon as possible"
Special Requirements	"basement needed", "solar-ready"

AI Personality:

- Bilingual (Urdu + English), switches on user preference
- Friendly, professional tone

- Asks one question at a time
- Handles vague answers: *"Don't worry, I'll help you estimate!"*

Recommendation Card (per result):

- Company name + verified badge
- Match score (e.g., 94%)
- Score breakdown: Price, Location, Rating, Style, Availability
- Why it matched: *"Budget fits / DHA experience / 4.8★rating"*
- CTA buttons: [View Profile] [Request Quote] [Compare]

7. Company Dashboard

Sections

Section	Description
 Incoming Requests	All client quote requests with status (New / Viewed / Responded / Accepted / Declined)
 Active Projects	Track ongoing projects with milestones and client info
 Material Marketplace	Browse and order from verified suppliers
 Profile Management	Edit portfolio, team, specializations, upload project photos
 Reviews	View and respond to client reviews
 Analytics	Leads received, response rate, profile views

Project Status Flow

New Request → Accepted → In Progress → Completed → Review Requested

Project Milestones

Each active project supports milestone tracking:

Foundation → Structure → Roofing → Finishing → Handover

Incoming Request Card

Client: Ahmed Raza

Budget: 1.2 - 1.5 Crore

Area: 10 Marla | Location: DHA Lahore

Timeline: 6 months

Message: "Looking for modern design..."

[Accept]

[Decline]

[Ask for Details]

8. Supplier Dashboard

Sections

Section	Description
 Product Listings	Add, edit, remove products with pricing + availability
 Inventory Tracker	Monitor stock levels with low-stock alerts
 Order Management	View and respond to purchase inquiries from companies
 Pricing Manager	Set bulk-pricing tiers, discount rules
 Sales Analytics	Order volume, top-selling products, revenue trends

Product Object (JSON)

json

```
{
  "id": "prod_001",
  "name": "OPC Cement 50kg",
  "category": "Cement",
  "pricePerUnit": 1050,
  "unit": "bag",
  "stock": 500,
  "minOrderQty": 10,
  "brand": "Lucky Cement",
  "deliveryAvailable": true,
  "location": "Lahore",
  "isActive": true
}
```

9. Admin Dashboard

9.1 Overview Analytics

Total Users	Companies	Suppliers	Requests
1,240	87	43	312
+12% ↑	Pending: 5	Pending: 2	Active: 94

9.2 User Management

- Paginated table: ID, Name, Email, Role, Status, Joined Date, Last Active
- Actions: View Profile, Suspend, Ban, Restore, Change Role
- Status badges: Active | Suspended | Banned | Pending
- Search by name or email; filter by role, status, date range

9.3 Company Verification Panel

Flow:

```
Submitted Application
↓
Admin Reviews Documents (License, NTN, Portfolio) – inline PDF viewer
↓
Approve → Verified Badge Added → Company Notified
Reject  → Reason Required    → Company Notified
```

- Filter by: Pending / Approved / Rejected / Flagged
- Admin can add internal notes per application

9.4 Supplier Verification Panel

Same flow as Company Verification, with supplier-specific documents (business registration, product catalog). Includes product listing review before approval.

9.5 Content Moderation

- Flagged reviews appear in a queue (user-reported)
- Admin actions: Approve, Remove, Escalate
- Fake/spam product listings: Remove with reason
- Response templates for notify-and-remove actions

9.6 Audit Logs

[2025-06-10 14:32] ADMIN: Approved company "BuildPro Lahore"

[2025-06-10 14:29] SYSTEM: New company signup – "Noor Builders"

[2025-06-10 14:15] CLIENT: Raza Ahmed submitted quote request #QR-2041

[2025-06-10 14:10] AI: Premium session started – User #U-882

Filterable by: User, Action Type, Date Range | Exportable as CSV (simulated for FYP)

9.7 Platform Monitoring

Log Type	What It Tracks
User Activity	Profile edits, account changes
Request Logs	Client → Company quote requests
AI Usage Logs	Premium chat sessions, token counts (simulated)
Moderation Logs	Admin actions (bans, removals)

10. AI Matchmaking Engine

10.1 Input Schema (JSON)

json

```
{
  "budget": 15000000,
  "area_marla": 10,
  "location": { "city": "Lahore", "area": "DHA Phase 5" },
  "style": "modern",
  "timeline_months": 6,
  "special_requirements": ["basement", "solar-ready"]
}
```

10.2 Scoring Algorithm

```
Final Score = Weighted Sum of:
price_match_score   × 0.35
location_score      × 0.25
rating_score        × 0.20
style_similarity     × 0.10
availability_score  × 0.10
```

Score Details:

Factor	Calculation
Price Match (35%)	$\boxed{1.0}$ if budget in range; penalized linearly if below; $\boxed{0.9}$ if budget exceeds (still good)
Location (25%)	Same city + same area = 1.0 / Same city only = 0.7 / Different city = 0.3
Rating (20%)	$\boxed{\text{company.avg_rating} / 5.0}$ — only companies with ≥ 3 reviews
Style (10%)	Keyword/tag matching: Modern↔Contemporary = 0.85, Modern↔Traditional = 0.30
Availability (10%)	0 active projects = 1.0 / $\leq 2 = 0.8$ / $\leq 5 = 0.5$ / $> 5 = 0.2$

10.3 Output Schema (JSON)

json

```

{
  "recommendations": [
    {
      "rank": 1,
      "company_id": "comp_001",
      "company_name": "BuildPro Lahore",
      "match_score": 94,
      "score_breakdown": {
        "price_match": 98,
        "location": 100,
        "rating": 88,
        "style": 90,
        "availability": 80
      },
      "explanation": "BuildPro Lahore's pricing fits your budget perfectly. They spe
      "verified": true,
      "portfolio_preview": ["img1.jpg", "img2.jpg"]
    }
  ],
  "generated_at": "2025-06-10T14:10:00Z",
  "requirements_used": {}
}

```

10.4 RAG Pipeline (Phase 3)

- **Vector Store:** ChromaDB (local) → Pinecone (production)
- **Embeddings:** OpenAI text-embedding-3-small or local Ollama model
- **Indexed documents:** Company profiles, project descriptions, reviews, material listings
- **Retrieval:** Top-K similarity search → reranked by scoring algorithm
- **LLM for explanation:** Claude API (for natural language explanation generation)

10.5 Mock AI (Phase 1 & 2)

- Hardcoded company data with metadata in JSON files
- Scoring algorithm runs entirely on frontend with dummy weights
- Fake "processing logs" with `setTimeout` delays for UX realism
- Bilingual responses (Urdu/English) pre-written for common question patterns

AI Analysis Log (simulated):

[10:42:01] Analyzing budget constraints...

[10:42:03] Filtering by location: Lahore

[10:42:05] Matching construction style: Modern

[10:42:07] Calculating compatibility scores...

[10:42:09] Top 5 companies identified ✓

11. Notification System

For FYP: All notifications are simulated in-app using Zustand state — no email/SMS/push.

Trigger	Recipient	Message
New quote request	Company	"Ahmed Raza requested a quote for 10 Marla house"
Quote responded	Client	"Bright Builders responded to your request"
Application approved	Company/Supplier	"Your account has been verified ✓"
Application rejected	Company/Supplier	"Your application was rejected. Reason: [X]"
New review received	Company	"You received a 5★ review from a client"
Low stock alert	Supplier	"OPC Cement stock below 50 bags"

In-App Messaging (Phase 2+):

- Client ↔ Company: Project discussion, file sharing
- Company ↔ Supplier: Material quotes, bulk order negotiation

12. Review & Trust System

Review Flow

Project Completed

↓

Client Receives Review Prompt (auto after project marked complete)

↓

Client Rates: Overall, Communication, Quality, Timeline

↓

Review Published (after spam check)

↓

Company Can Respond Publicly (once only)

Trust Signals on Company Profile

-  Verified Badge (admin-approved)
-  Aggregate Rating (weighted, recent reviews weighted more)
-  Completed Projects Count
-  Member Since
-  Response Rate %

Anti-Fraud Rules

- Reviews only from clients who completed a project with that company
 - One review per project
 - Admin can flag/remove reviews with notes
 - Spam filter for keyword stuffing (simulated for FYP)
-

13. Frontend Architecture

Folder Structure

```
src/
├── components/
│   ├── common/           # Button, Card, Badge, Modal, Input, Avatar, Toast
│   ├── layout/           # Navbar, Sidebar, Footer
│   ├── client/           # ClientDashboard, AIChat, CompanyCard, CompareTable
│   ├── company/          # CompanyDashboard, RequestCard, ProjectCard, Milestones
│   ├── supplier/         # SupplierDashboard, ProductCard, InventoryTable
│   └── admin/            # AdminDashboard, UserTable, VerificationPanel, LogStream
├── pages/
│   ├── Landing.jsx
│   ├── Login.jsx         # (mock - sets role in Zustand, no real auth)
│   ├── Signup.jsx        # (mock - simulates pending state for company/supplier)
│   ├── client/
│   ├── company/
│   ├── supplier/
│   └── admin/
├── store/                # Zustand stores (auth, requests, notifications, products)
├── hooks/                # Custom React hooks
├── utils/                # Helpers, formatters, scoring algorithm
├── data/                 # JSON data files (companies, suppliers, requests, users)
└── styles/               # Tailwind config, global CSS
```

State Management (Zustand)

```
javascript

// Auth Store (mock – role set on login, persisted in localStorage)
const useAuthStore = create((set) => ({
  user: null,
  role: null,
  login: (userData) => set({ user: userData, role: userData.role }),
  logout: () => set({ user: null, role: null }),
}));

// Notifications Store
const useNotificationStore = create((set) => ({
  notifications: [],
  addNotification: (n) => set((s) => ({ notifications: [n, ...s.notifications] })),
  markRead: (id) => set((s) => ({
    notifications: s.notifications.map(n => n.id === id ? { ...n, read: true } : n)
  })),
}));
```

Route Protection

```
javascript

// ProtectedRoute – reads role from Zustand/localStorage
<Route path="/client/*" element={
  <ProtectedRoute allowedRoles={["client"]} >
    <ClientDashboard />
  </ProtectedRoute>
} />
```

Reusable Components

Component	Usage
<StatusBadge>	Pending / Approved / Rejected / Banned
<MatchScoreRing>	Circular progress showing AI match %
<VerifiedBadge>	Gold checkmark on company profile
<LogStream>	AI analysis logs panel (scrollable)
<CompanyCard>	Browse, compare, and recommendation panel
<RequestCard>	Company dashboard incoming requests

Component	Usage
<Skeleton>	Loading states for all card types

14. Page Inventory (MVP)

Shared Pages

Page	Path
Landing Page	/
Login (mock)	/login
Signup (mock)	/signup
404 Not Found	*

Client Pages

Page	Path
Dashboard	/client/dashboard
Browse Companies	/client/companies
Company Detail	/client/companies/:id
Compare Companies	/client/compare
AI Chat (Premium)	/client/ai-chat
Premium Upgrade	/client/upgrade
My Requests	/client/requests
Saved Companies	/client/saved
My Reviews	/client/reviews
Profile Settings	/client/settings

Company Pages

Page	Path
Dashboard	/company/dashboard
Incoming Requests	/company/requests
Active Projects	/company/projects
Project Detail	/company/projects/:id
Material Marketplace	/company/marketplace
Public Profile Builder	/company/profile
Settings	/company/settings

Supplier Pages

Page	Path
Dashboard	/supplier/dashboard
Product Management	/supplier/products
Inventory	/supplier/inventory
Orders	/supplier/orders
Profile	/supplier/profile
Settings	/supplier/settings

Admin Pages

Page	Path
Dashboard	/admin/dashboard
User Management	/admin/users
Company Verification	/admin/verify/companies
Supplier Verification	/admin/verify/suppliers
Content Moderation	/admin/moderation

Page	Path
Activity Logs	/admin/logs
Analytics	/admin/analytics

15. JSON Data Schema

┆ All data lives in `src/data/` as JSON files. No database or backend for FYP.

users.json

json

```
[
  {
    "id": "usr_001",
    "name": "Ahmed Raza",
    "email": "ahmed@example.com",
    "role": "client",
    "isPremium": false,
    "isVerified": true,
    "status": "active",
    "createdAt": "2025-01-10T08:00:00Z"
  },
  {
    "id": "usr_002",
    "name": "Hania Khan",
    "email": "hania@example.com",
    "role": "client",
    "isPremium": true,
    "isVerified": true,
    "status": "active",
    "createdAt": "2025-02-05T10:00:00Z"
  }
]
```

companies.json

json

```
[
  {
    "id": "comp_001",
    "userId": "usr_010",
    "name": "Bright Builders",
    "city": "Lahore",
    "area": "DHA",
    "bio": "Premium construction specialists with 15+ years in DHA.",
    "specialization": ["Modern", "Commercial"],
    "minPrice": 100000000,
    "maxPrice": 200000000,
    "rating": 4.8,
    "reviewCount": 32,
    "projectsCompleted": 47,
    "activeProjects": 2,
    "verified": true,
    "status": "approved",
    "teamSize": 45,
    "yearsInBusiness": 12,
    "portfolioImages": ["proj1.jpg", "proj2.jpg"]
  },
  {
    "id": "comp_002",
    "userId": "usr_011",
    "name": "Al-Rehman Constructors",
    "city": "Lahore",
    "area": "Bahria Town",
    "bio": "Trusted residential builders serving Bahria Town since 2008.",
    "specialization": ["Traditional", "Residential"],
    "minPrice": 80000000,
    "maxPrice": 140000000,
    "rating": 4.5,
    "reviewCount": 58,
    "projectsCompleted": 82,
    "activeProjects": 1,
    "verified": true,
    "status": "approved",
    "teamSize": 30,
    "yearsInBusiness": 16,
    "portfolioImages": []
  }
]
```

suppliers.json

json

```
[
  {
    "id": "sup_001",
    "userId": "usr_020",
    "businessName": "Tariq Cement Depot",
    "location": "Lahore",
    "categories": ["Cement", "Bricks"],
    "verified": true,
    "status": "approved"
  }
]
```

products.json

json

```
[
  {
    "id": "prod_001",
    "supplierId": "sup_001",
    "name": "OPC Cement 50kg",
    "category": "Cement",
    "pricePerUnit": 1050,
    "unit": "bag",
    "stock": 500,
    "minOrderQty": 10,
    "brand": "Lucky Cement",
    "deliveryAvailable": true,
    "isActive": true
  }
]
```

requests.json

json

```
[
  {
    "id": "req_001",
    "clientId": "usr_001",
    "companyId": "comp_001",
    "budget": 15000000,
    "area": 10,
    "location": "DHA Lahore",
    "style": "Modern",
    "timeline": 6,
    "message": "Looking for a 10 marla modern house with basement.",
    "status": "pending",
    "createdAt": "2025-06-10T09:00:00Z"
  }
]
```

projects.json

```
json

[
  {
    "id": "proj_001",
    "companyId": "comp_001",
    "clientId": "usr_001",
    "requestId": "req_001",
    "title": "10 Marla Modern House - DHA Phase 5",
    "status": "in_progress",
    "currentMilestone": "Structure",
    "milestones": ["Foundation", "Structure", "Roofing", "Finishing", "Handover"],
    "startDate": "2025-03-01",
    "estimatedEndDate": "2025-09-01"
  }
]
```

reviews.json

```
json
```

```
[
  {
    "id": "rev_001",
    "projectId": "proj_001",
    "clientId": "usr_001",
    "companyId": "comp_001",
    "ratingOverall": 5,
    "ratingQuality": 5,
    "ratingCommunication": 4,
    "ratingTimeline": 5,
    "body": "Excellent work, very professional team.",
    "isFlagged": false,
    "isVisible": true,
    "createdAt": "2025-09-05T12:00:00Z"
  }
]
```

notifications.json

json

```
[
  {
    "id": "notif_001",
    "userId": "usr_010",
    "type": "new_request",
    "message": "Ahmed Raza requested a quote for 10 Marla house",
    "read": false,
    "createdAt": "2025-06-10T09:05:00Z"
  }
]
```

ai_sessions.json

json

```
[
  {
    "id": "sess_001",
    "clientId": "usr_002",
    "requirements": {
      "budget": 15000000,
      "area_marla": 10,
      "location": "DHA Lahore",
      "style": "Modern",
      "timeline_months": 6,
      "special": ["basement", "solar-ready"]
    },
    "topCompanies": ["comp_001", "comp_002"],
    "createdAt": "2025-06-10T14:10:00Z"
  }
]
```

admin_logs.json

```
json

[
  {
    "id": "log_001",
    "adminId": "usr_admin",
    "actionType": "company_approved",
    "targetId": "comp_001",
    "targetType": "company",
    "reason": "All documents verified",
    "createdAt": "2025-06-10T14:32:00Z"
  }
]
```

16. Development Phases

Phase 1 — Static UI (Weeks 1-3)

Goal: All pages look great, navigation works, JSON dummy data renders.

- Landing page with hero, features, how-it-works sections
- Login / Signup forms with role selection (mock — sets Zustand state)
- Client Normal Mode: Browse + Filter + Compare
- Client Premium Mode: Split-screen AI chat UI + dummy recommendations

- Company Dashboard with dummy requests and projects
- Supplier Dashboard with dummy product listings
- Admin Dashboard with tables and status badges
- Responsive layout (mobile, tablet, desktop)

Deliverable: Fully navigable static React app with no network calls.

Phase 2 — Routing & State (Weeks 4-5)

Goal: App behaves like a real product with persistent state.

- React Router setup with role-based protected routes
- Zustand stores (auth, requests, notifications)
- Role-based redirects on login
- Request flow: client submits → appears in company dashboard (via Zustand)
- Admin actions: approve/reject → status updates across app (via Zustand)
- Notification state simulation
- AI chat state machine (multi-step requirement collection)
- Persistent mock data via `localStorage`

Deliverable: Working multi-role app with realistic interactions, no backend.

Phase 3 — AI Integration (Weeks 6-9)

Goal: Real AI chat integration (API call to Claude/OpenAI).

- Real AI chat (Claude API / OpenAI API)
- RAG pipeline for matchmaking (LangChain + ChromaDB local)
- Scoring algorithm fully implemented in frontend utilities
- File uploads for company docs (handled client-side, stored as base64 in JSON)

Deliverable: AI-powered matchmaking working on localhost.

Phase 4 — Polish & Demo (Weeks 10-11)

Goal: Demo-ready FYP submission.

- Responsive polish on all screens
- Loading states, error states, empty states
- Framer Motion animations (page transitions, card hover, AI log stream)
- Presentation slides and demo video
- Documentation finalization + README

Deliverable: Submission-ready FYP.

17. Testing Strategy

Unit Testing (Jest + React Testing Library)

- Individual components (Card, Button, Form inputs)
- Utility functions (score calculator, formatters)
- Zustand store actions

Integration Testing (Cypress or Playwright)

- Login → role redirect flow
- Client submits request → appears in company dashboard
- Admin approves company → verified badge appears

Manual Functional Testing Checklist

Scenario	Expected Result
Client selects Normal mode	Redirected to browse/filter view
Client activates Premium	AI chat UI loads with Sara
AI collects all requirements	Recommendations appear on right panel
Company submits signup	Status = Pending, read-only dashboard
Admin approves company	Company status = Verified, badge added
Admin bans user	User role blocked from dashboard
Supplier adds product	Appears in company marketplace

18. UI/UX Design System

Design Principles

1. **Clarity first** — Users should never be confused about where they are or what to do next
2. **Role-aware UI** — Each role has a distinct color accent and layout identity
3. **Mobile-responsive** — All layouts work on 375px and above
4. **Progressive disclosure** — Show complexity only when needed (especially Premium AI)

Color Palette

Token	Hex	Usage
<code>--primary</code>	 #1D4ED8	Buttons, links, highlights
<code>--primary-dark</code>	 #1E3A8A	Hover states
<code>--client-accent</code>	 #10B981	Client UI accents (Emerald)
<code>--company-accent</code>	 #4F46E5	Company UI accents (Indigo)
<code>--supplier-accent</code>	 #F59E0B	Supplier UI accents (Amber)
<code>--admin-accent</code>	 #DC2626	Admin UI accents (Red)
<code>--premium</code>	 #FBBF24	Premium badge, gold indicators
<code>--success</code>	 #22C55E	Verified badges, success states
<code>--danger</code>	 #EF4444	Errors, bans, rejections
<code>--warning</code>	 #EAB308	Pending states, alerts
<code>--bg</code>	 #F8FAFC	Page background
<code>--text-primary</code>	 #111827	Primary text

Typography

Style	Font	Size	Weight
Display / Page Title	Inter	36px	700
Section Heading	Inter	28px	600
Card Title	Inter	18px	600
Body	Inter	16px	400
Small / Caption	Inter	13px	400
Label	Inter	12px	500

Status Badges

-  **Verified** — green, checkmark icon

-  **Pending** — amber, clock icon
-  **Rejected** — red, X icon
-  **Premium** — purple gradient, diamond icon
-  **Active** — blue, dot icon
-  **Suspended** — gray, pause icon

Animations (Framer Motion)

Trigger	Animation
Page transition	Fade + slide up (0.3s)
Card hover	Scale 1.02 + shadow increase
AI log entry	Slide in from left (staggered)
Recommendation card	Fade + scale in (staggered delay)
Modal open	Scale from 0.9 + fade
Score bar	Width animate from 0% on enter

Responsive Breakpoints

Breakpoint	Width	Layout
Mobile	< 640px	Single column; collapsed sidebar
Tablet	640-1024px	2-column grid; slide-out sidebar
Desktop	> 1024px	Full layout; persistent sidebar

19. Out of Scope

The following are **not** part of this FYP:

Feature	Reason
Payment Processing	Requires PCI compliance, banking integration
Cloud Deployment	FYP runs on localhost only
Mobile App (iOS/Android)	Separate future project
Real-time Chat (WebSockets)	Post-FYP with Socket.io
Email / SMS Notifications	Simulated in-app only
Third-party Map Integration	Cost; mockable for FYP
Multi-language CMS	Content is hardcoded for FYP
Legal Contracts / Digital Signatures	Out of scope
Real Database (PostgreSQL)	JSON files used instead for FYP

20. Feature Comparison Matrix

Feature	Normal Client	Premium Client	Company	Supplier	Admin
Browse Companies	✓	✓	—	—	✓
Filter & Compare	✓	✓	—	—	—
Request Quote	✓	✓	—	—	—
AI Chat (Sara)	✗	✓	—	—	—
AI Recommendations	✗	✓	—	—	—
View Reviews	✓	✓	✓	—	✓
Write Review	✓ (post-project)	✓	—	—	—
Manage Projects	—	—	✓	—	—
Marketplace Access	—	—	✓	—	—
Product Listings	—	—	—	✓	—
Order Management	—	—	—	✓	—
Approve Users	—	—	—	—	✓

Feature	Normal Client	Premium Client	Company	Supplier	Admin
View All Logs	—	—	—	—	✓
Ban / Suspend Users	—	—	—	—	✓
Platform Analytics	—	—	—	—	✓

21. Key Differentiators

Aspect	Regular Directory	Smart Construction Connect
Discovery	Manual search	AI-powered matchmaking
Language	English only	Urdu + English
Trust	No verification	Admin-verified + reviews
Matching	Generic	Score-based with explanation
Supplier link	Not present	Integrated marketplace
Governance	None	Full admin control layer
Market	Generic	Pakistan-specific (PKR, Marla, DHA)

"The only platform in Pakistan where a homeowner can describe their dream house in Urdu, and receive AI-matched, verified, scored construction companies in under 60 seconds."

22. Risk Register

Risk	Likelihood	Impact	Mitigation
AI API rate limits in demo	Medium	High	Pre-generate responses for demo scenarios
LLM gives wrong recommendations	Medium	Medium	Scoring algorithm overrides LLM; show breakdown
Mock data looks unrealistic	Low	Medium	Use real Pakistani company names and Lahore locations
Phase 3 AI not complete for demo	Medium	High	Ensure Phase 1-2 are demo-ready as fallback

Risk	Likelihood	Impact	Mitigation
Urdu AI responses broken	Low	High	Test all Urdu inputs; hardcode fallback responses
Admin panel complexity causes delays	Low	Medium	Build admin last; core flows take priority

23. Glossary

Term	Definition
RAG	Retrieval-Augmented Generation — AI technique that retrieves relevant documents before generating a response
Match Score	A weighted percentage indicating how well a company fits a client's requirements
Verified Badge	Visual indicator that a company/supplier has been approved by Admin
Premium Mode	AI-powered client mode with chatbot and recommendation features
Marla	Pakistani unit of area; 1 Marla ≈ 25.3 m ²
NTN	National Tax Number (Pakistani business registration)
RBAC	Role-Based Access Control — restricts system access based on user role
Kanban	Visual project management framework with columns representing stages
MOQ	Minimum Order Quantity — minimum units a supplier will sell
DHA	Defence Housing Authority — premium residential area in Pakistani cities